Introduction to Software Engineering

Project Idea & Team Contract



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Project Idea & Team Contract

Objectives

This document focus on the following topics:

- ✔ Describing the project idea at a high level so the teaching staff can approve it
- ✔ Agreeing on how the team will work together (team contract)
- ✔ PA0 is not graded, but it is required before the team starts PA1

1 Team Information

Item	Value
Team ID	02
Team name	BDD
Team lead	Đoàn Quốc Bảo
Communication channel	Messenger group chat

Student ID	Full name	Email	Main role	Backup role
24127326	Đoàn Quốc Bảo	dqbao2432@clc.fitus.edu.vn	Team lead / PM	Developer
24127025	Lê Minh Đức	lmduc2433@clc.fitus.edu.vn	Business analyst	Designer
24127003	Vũ Trần Minh Hiếu	vtmhieu2416@clc.fitus.edu.vn	Designer	Developer
24127173	Nguyễn Tuấn Hùng	nthung2432@clc.fitus.edu.vn	Developer	Tester
24127008	Trương Nhật Phát	tnphat2418@clc.fitus.edu.vn	Tester	PM / Business analyst

2 Project Idea Overview

2.1 Project Name and One-Sentence Pitch

Portal67 - a web app that improves upon the current HCMUS Portal system.

2.2 Problem and Motivation

The system provides additional features that help solve students' and teachers' pain points when using the current Portal system. The pain points consist of insufficient GPA tracking, timetable planning, and a slow course registration experience due to server overload. Currently, there are several solutions provided by the student community, such as KHTN GPA 2.0 extension, etc. The majority of them are effective at solving their specific goals, but they have not yet covered all of the mentioned pain points.

2.3 Target Users

User group	What they need from the system
Students (both undergrads and postgrads)	Robust course registration system. Expect 99.99% operational uptime with a modern interface
Teachers	Simplified system for course & scoring management
Administrative worker	Unified resource management and document approval platform. Replace all existing 3rd-party hosted poll & form systems

2.4 Main Features

#	Feature	Short description	Must-have?
1	Course & Specialized topic registration	Browse offered sections and special topics; see seats and schedules; submit/cancel	Yes

		registration. Validate basic seat limits and time conflicts using seeded records.	
2	Tuition management	View semester charges, payment records, balances, and payment status. Staff can update simulated payment records; no money is transferred.	Yes
3	Academic result management	Lecturers enter demo grades; students view course results and an estimated semester/cumulative GPA computed from configurable sample rules.	Yes
4	Course & Exam scheduling	Show a student's weekly course timetable and exam list; display overlapping sessions and registration conflicts.	Yes
5	Resource Management	Longer-term room/equipment availability and booking for university staff. Deferred beyond the MVP unless core work finishes early.	No
6	Document Request & Approval	Students submit a simple document request; staff approve/reject with comments; students view status and history. No legally valid document is issued.	Yes

2.5 Similar Products

Currently, there is the HCMUS Portal system and Google Classroom. The Portal system for CTĐA and Regular separately operated. Other management works often stretch across various platforms, like the Moodle system.

2.6 Platform and Technology Ideas

- Platform: Website
- Programming language: Typescript, Golang
- Technology stack:
    - Frontend: React Router, Svelte, TailwindCSS
    - Backend: NestJS, Golang
    - CI Orchestration: Kubernetes, Jenkins, Github Action
    - Database: PostgreSQL + Redis
    - Version Control: GitHub
    - Web server, load balancer, reverse proxy: Nginx
    - Message broker: Apache Kafka

2.7 Feasibility, Scope and Open Questions

- Yes, team can build the Must-have features (Features 1, 2, 3, 4, and 6) within the 3-week timeline (Weeks 8–10)

Open question:

- We wonder if HCMUS (or at least FIT) would actually replace their own Portal system to use our system if we score a 10 on this.

3 Team Contract

3.1 Goals and Expectations

The team is aiming for a 9+ score. Each member is willing to spend at least 6 hours per week.

3.2 Communication

- ✔ Main channel for discussion: Messenger.
- ✔ Responses to messages must be within 24 hours on weekdays.
- ✔ Decisions and files are kept in a shared drive.

3.3 Meetings

- ✔ Internal team meetings: Online, Saturday or Sunday weekly (the specific meeting time or any change will be announced at least 3 days in advance of each meeting).
- ✔ Attendance rule: inform the team in advance if you cannot attend; missed meetings are recorded in the weekly status report, and counted as a strike.
- ✔ Weekly status report deadline chosen by the team: Sunday 23h59

3.4 Roles and Responsibilities

Extra responsibilities (presentation, etc) are split equally, decided by the team leader.

3.5 Work Standards

- ✔ Each member must finish work before the deadline set by the team leader. If not satisfied with the deadline, communicate directly with the team leader for a final decision.
- ✔ Quality: every document section and pull request is reviewed by at least one other member.
- ✔ Tools: every task is on the board with an owner and due date; code goes through pull requests.
- ✔ Code management: These actions are prohibited: Unnecessary, non-PR push on the main branch; Force pushing a branch with two or more contributors. If caught, a fine of 20.000VNĐ shall be issued internally to the individual performed

3.6 Use of AI Tools

Members are required to write how AI was used. Agent usage needs strict human approval.

3.7 Decision Making

Majority voting (passed if >50%). In case of a tie, the team leader will decide.

3.8 Handling Conflicts and Missed Work

- ✔ Conflicts and missed work are handled via communication. Unresolved conflicts that severely affect the development process will be reported to the TA. The members who miss a deadline will have a 24-hour grace period to give a valid reason.
- ✔ Missed work or meetings will be counted as strikes, a valid reason will not result in a strike

  Strike 1: A new deadline, and a record in the weekly status report.

  Strike 2: Inform the TA, the overall individual contribution factor will be affected.

  Strike 3: Inform the TA, the member will be asked to be removed from the group.

3.9 Signatures

Student ID	Full name	Signature	Date
24127326	Đoàn Quốc Bảo	(signed)	14/10/2026
24127025	Lê Minh Đức	(signed)	14/10/2026
24127003	Vũ Trần Minh Hiếu	(signed)	14/10/2026
24127173	Nguyễn Tuấn Hùng	(signed)	14/10/2026
24127008	Trương Nhật Phát	(signed)	14/10/2026

4 AI Usage Log

No AI tools were used in this document.